# MSG No. 44.234 $I_amm2$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: `4a`, site symmetry: `mm2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5,6,7,8]` |
| 3 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[9,10,11,12]` |
| 4 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[13,14,15,16]` |

Table 2: Wyckoff site: `4b`, site symmetry: `m'm2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ 0,\ z\right]$ | `[1,4,10,11]` |
| 2 | $\left[\frac{3}{4},\ 0,\ z\right]$ | `[2,3,9,12]` |
| 3 | $\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5,8,14,15]` |
| 4 | $\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[6,7,13,16]` |

Table 3: Wyckoff site: `8c`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | `[1,4]` |
| 2 | $[-x,\ 0,\ z]$ | `[2,3]` |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5,8]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[6,7]` |
| 5 | $\left[x+\frac{1}{2},\ 0,\ z\right]$ | `[9,12]` |
| 6 | $\left[\frac{1}{2}-x,\ 0,\ z\right]$ | `[10,11]` |
| 7 | $\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[13,16]` |
| 8 | $\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[14,15]` |

Table 4: Wyckoff site: `8d`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | `[1,3]` |
| 2 | $[0,\ -y,\ z]$ | `[2,4]` |
| 3 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5,7]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[6,8]` |
| 5 | $\left[\frac{1}{2},\ y,\ z\right]$ | `[9,11]` |
| 6 | $\left[\frac{1}{2},\ -y,\ z\right]$ | `[10,12]` |
| 7 | $\left[0,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[13,15]` |
| 8 | $\left[0,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[14,16]` |

Table 5: Wyckoff site: `8e`, site symmetry: `m'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ z\right]$ | `[1,11]` |
| 2 | $\left[\frac{3}{4},\ -y,\ z\right]$ | `[2,12]` |
| 3 | $\left[\frac{3}{4},\ y,\ z\right]$ | `[3,9]` |
| 4 | $\left[\frac{1}{4},\ -y,\ z\right]$ | `[4,10]` |
| 5 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5,15]` |
| 6 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[6,16]` |
| 7 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[7,13]` |
| 8 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[8,14]` |

Table 6: Wyckoff site: `16f`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[-x,\ -y,\ z]$ | `[2]` |
| 3 | $[-x,\ y,\ z]$ | `[3]` |
| 4 | $[x,\ -y,\ z]$ | `[4]` |
| 5 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[6]` |
| 7 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[8]` |
| 9 | $\left[x+\frac{1}{2},\ y,\ z\right]$ | `[9]` |
| 10 | $\left[\frac{1}{2}-x,\ -y,\ z\right]$ | `[10]` |
| 11 | $\left[\frac{1}{2}-x,\ y,\ z\right]$ | `[11]` |
| 12 | $\left[x+\frac{1}{2},\ -y,\ z\right]$ | `[12]` |
| 13 | $\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[13]` |
| 14 | $\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[14]` |
| 15 | $\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[15]` |
| 16 | $\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[16]` |